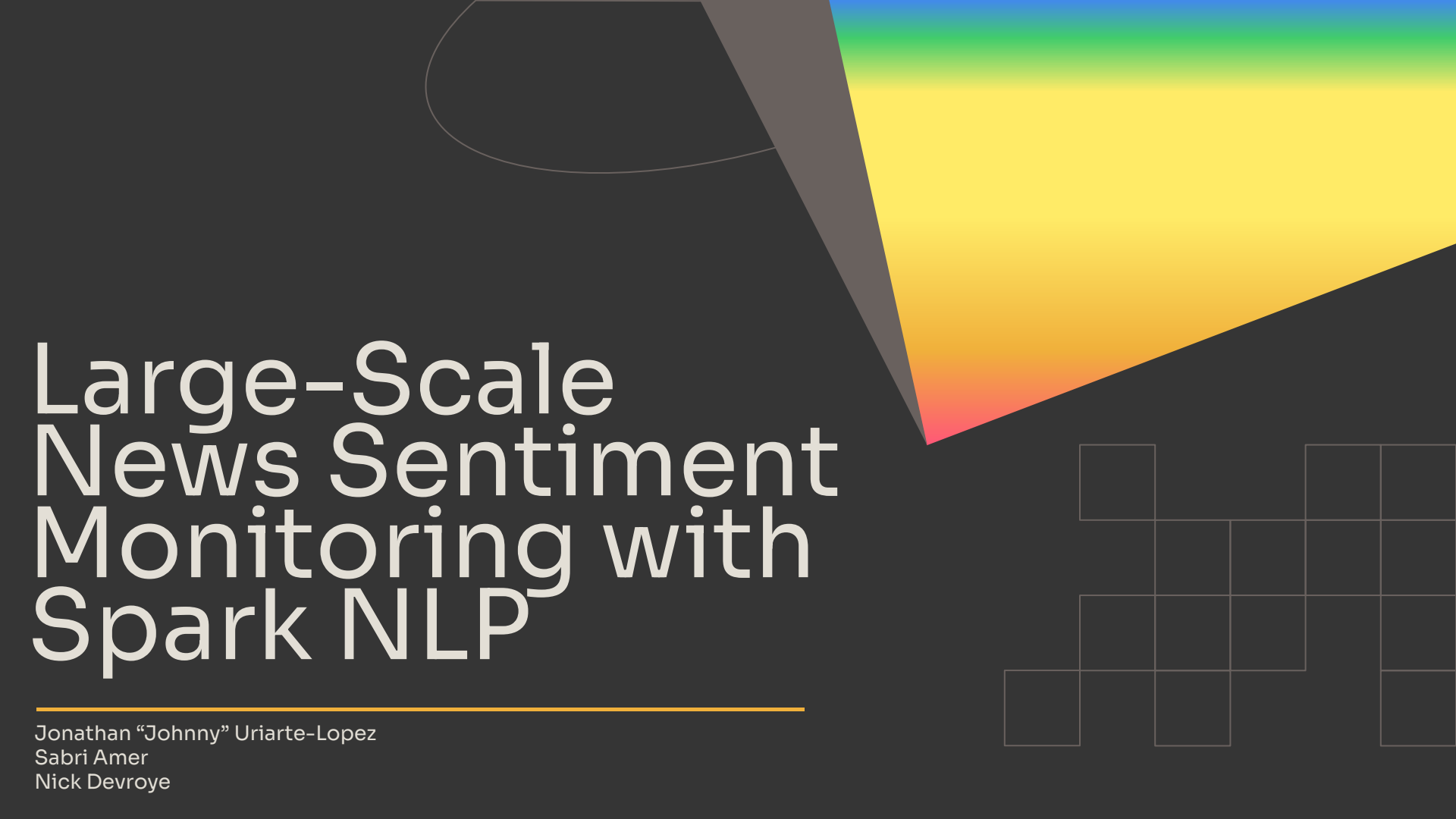




Large-Scale News Sentiment Monitoring with Spark NLP

Jonathan “Johnny” Uriarte-Lopez
Sabri Amer
Nick Devroye



Jonathan “Johnny”
Uriarte-Lopez



Sabri Amer
(Presenting via
YouTube)



Nick Devroye

Meet The Team :D

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BREAKING NEWS WITH SABRI!

(insert link when ready)



[Spongebob fish link](#)



Introduction

Sentiment monitoring of global news is critical for government, businesses, and security organizations.

To understand:

- Public opinion.
- Emerging crises.
- Disinformation campaigns.

Manual analysis of multilingual news published daily is impossible.

Big data engineering tools are needed:

- Natural Language Processing (NLP).
- Scalable Extract, Transform, and Load (ETL) workflows.
- Distributed analytics



[Superman logo wiki](https://en.wikipedia.org/wiki/Superman)



https://en.wikipedia.org/wiki/Daily_Planet

Significance and Objectives



[International Image Link](#)

Significance:

- Reviewing the real-time news reportings from across the globe and being able to quickly summarize the general impact could be a valuable tool for anyone who needs to stay on top of global events and react quickly to their impacts.
 - Government workers, Foreign Affairs, News Executives
- This project would will look at positive, negative, and neutral outcomes on technology journal articles with Spark NLP.

Significance and Objectives cont.



- Ingest and preprocess more than 10 GB of global news data using Hadoop/Spark.
- Apply Spark NLP sentiment analysis models to classify emotional tone (positive, negative, neutral) of the news report.
- Showcase analysis using Python Matplotlib package for visualization.
- This project explored technology trends during May 2018.
- End product is understanding trends in technology.

[Global News Image Link](#)

Data Sources

Global Database of Events, Language, and Tone (GDELT)

Global database of worldwide news:

- <https://www.gdeltproject.org/data.html>

This project is supported by [Google Jigsaw](#).

- Technology incubator within Google.
- Goal is to empower groups and individuals.

Up to date on current news, and historical archives all the way back to 1979.

Data Collection

Three ways to use GDELT data:

1. GDELT API.
2. Google BigQuery.
3. Download raw files.
 - a. Sorted by year_month.
 - b. As .csv.zip files.

We ended up downloading raw files using python web scraping techniques.

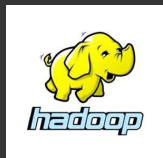


GDELT 3.0

Tools & Technologies

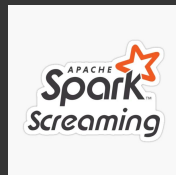
End Result Architecture

- The raw news data will be stored into a distributed file system with HDFS.
 - Downloaded as zip files and later converted to parquet files.
- Apache Spark Streaming Used for Batch Analysis.
 - Enables simulation of real time “events” and has native compatibility with the hadoop ecosystem from the parquet files.
- Sentiment Analysis will be conducted with Spark NLP
 - End result was a simple categorization, but more complex NLP models can be used.
- Data stored locally, analyzed with Spark SQL.
- Visuals created with Matplotlib python package.
 - Workaround since we could not get Hive and Hue working.



Hadoop HDFS

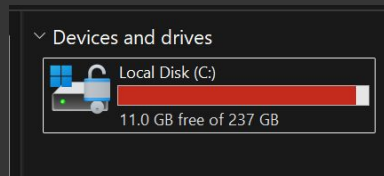
Raw Data Storage
(JSON/CSV/Parquet)



Spark Streaming
Real-time Streaming
Simulation



Spark NLP
Real-Time Sentiment
Analysis



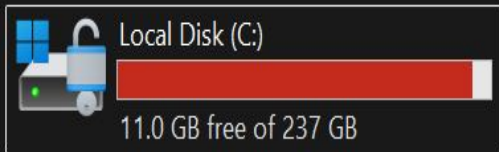
Analytics Stored
Locally



Matplotlib
Proof of Concept
Visuals

Extra Extra! Reports, Insights, and Recommendation

Devices and drives

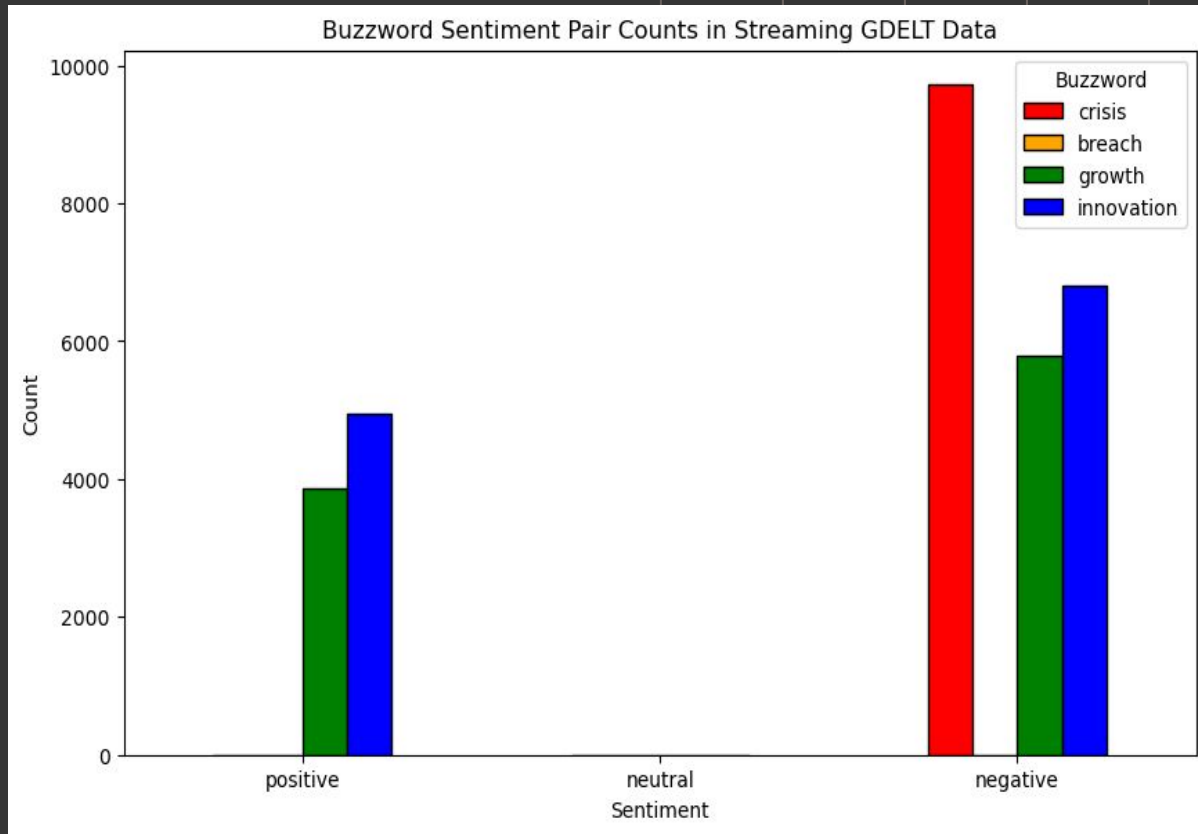


GDEL File Size & Usage

Data only contains metadata, not the article themselves.

Zip files contain url hyperlink to article.

Metadata useful for sentiment analysis.



Future Goals & Challenges

Memory Management Improvements

Delete temporary zip files after parquet is created.

- Save space on disk.

Disk space will be an ongoing issue

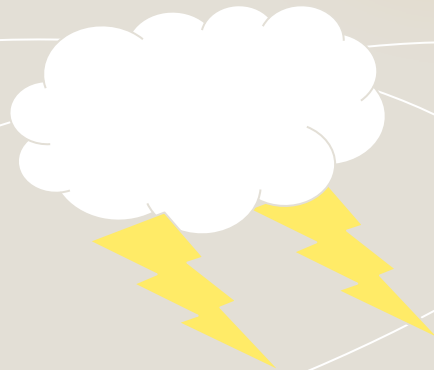
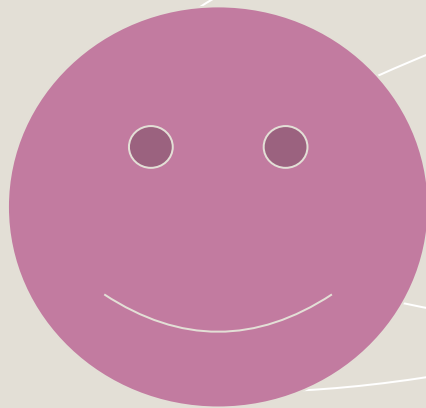
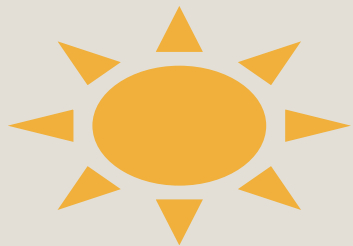
- Do not underestimate size of GDELP data!

Business Goals

Get a temperature check on ongoing events in the market.

- Act upon any profitable trends (need for a product or service)
- Or defend against potential liability (hacker attacks and their techniques).

DEMO!!!



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